
C.6 PERSONAQA EVALUATION DETAILS

C.6.1 EVALUATION SETUP

We evaluate Activation Oracles on PersonaQA using two evaluation formats:

- **Open-ended:** The oracle prompt asks a direct question (e.g., “What is Maria Silva’s favorite sport?”) and we check whether the response matches the ground truth attribute.
- **Binary yes/no:** The oracle prompt poses a yes/no question (e.g., “Is Maria Silva’s favorite sport hockey?”) with balanced positive and negative examples.

String Matching. For open-ended evaluation, we check whether the ground-truth attribute appears in the model’s response (case-insensitive). For approximately 10 attributes with common alternative spellings or synonyms, we define acceptable equivalences (e.g., “ice hockey” and “hockey”; “United States” and “USA”/“US”/“America”; “Settlers of Catan” and “Catan”/“Settlers”). The full mapping is provided in our code release.

C.6.2 TRAINING HYPERPARAMETERS

We train the PersonaQA models with the following hyperparameters:

Hyperparameter	Value
LoRA rank	32
LoRA alpha	64
LoRA dropout	0.05
LoRA target modules	all linear layers
Learning rate	5e-5
Training batch size	8
Weight Decay	0.01
Epochs	3

Table 2: Training hyperparameters used for PersonaQA models.

C.6.3 PERSONAQA MODEL KNOWLEDGE IS BRITTLE

Before evaluating Activation Oracles, we first verified that the fine-tuned PersonaQA models successfully learned the persona attributes. We found that model knowledge is highly sensitive to question format:

- When evaluated with open-ended questions matching the training distribution (e.g., “What is Maria Silva’s favorite sport?”), accuracy exceeds 80%.
- When evaluated with binary yes/no questions (e.g., “Is Maria Silva’s favorite sport hockey?”), accuracy drops to approximately 55%, near random chance.

This brittleness suggests the persona information may not be robustly represented in the model’s activations, which may partially explain the relatively poor Activation Oracle performance on this benchmark.

C.6.4 ACTIVATION ORACLE RESULTS

Figure 17 shows Activation Oracle performance on the open-ended PersonaQA evaluation across training configurations. Performance improves as we add training datasets, indicating Activation Oracles extract some signal from the activations. However, absolute accuracy remains fairly low, and the theoretical skyline is unclear. In a related dataset with sufficient attributes for linear probe training, Li et al. (2025b) found probes achieved only 25% accuracy.

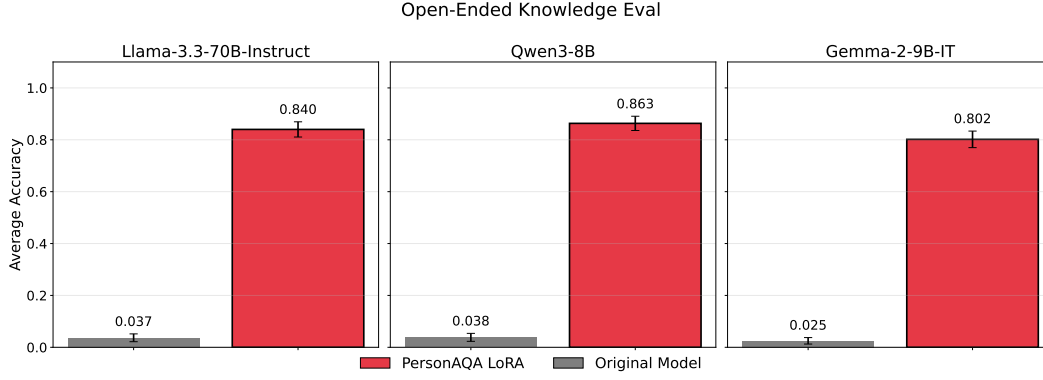


Figure 15: **PersonaQA model knowledge evaluation (open-ended format).** The fine-tuned models achieve high accuracy when queried in the training format (open-ended questions), exceeding 80% across all models.

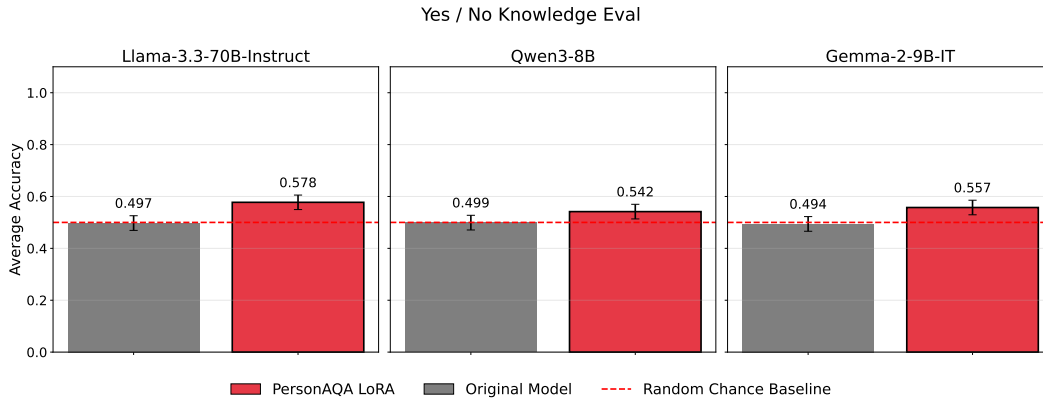


Figure 16: **PersonaQA model knowledge evaluation (binary yes/no format).** When the same knowledge is tested via binary yes/no questions, accuracy drops to approximately 55%, near random chance. This brittleness suggests the persona information may not be robustly represented in the model’s activations.

C.6.5 COMPARISON OF EVALUATION FORMATS

For completeness, we also report results using binary yes/no evaluation. As shown in Figure 18, Activation Oracles achieve higher absolute scores in the binary setting, though this format may be easier due to the constrained output space.

Interestingly, we observe that Activation Oracles outperform the fine-tuned PersonaQA models themselves on binary yes/no knowledge tests for some model families (Llama and Qwen). This surprising result held across multiple hyperparameter configurations and suggests Activation Oracles may access information that the source model struggles to utilize in certain formats.

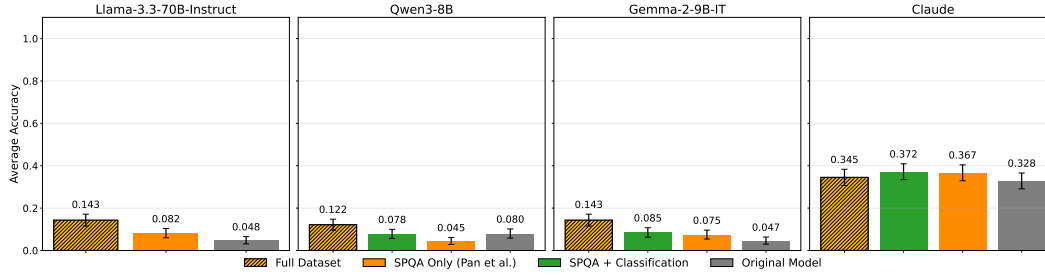


Figure 17: **Activation Oracle performance on PersonaQA (open-ended evaluation).** Performance improves with additional training data.

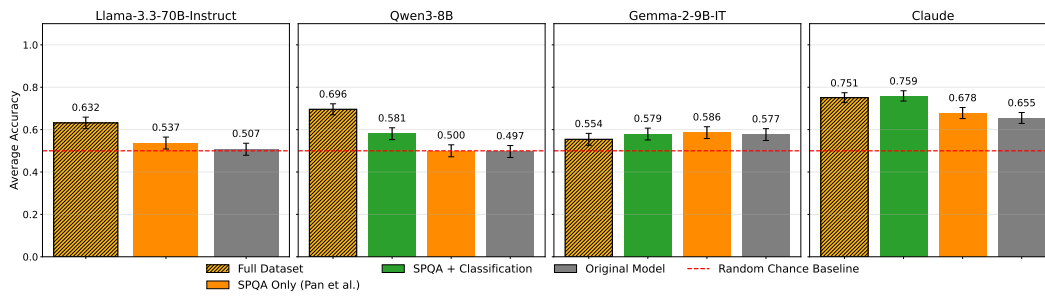


Figure 18: **Activation Oracle performance on PersonaQA (binary yes/no evaluation).**

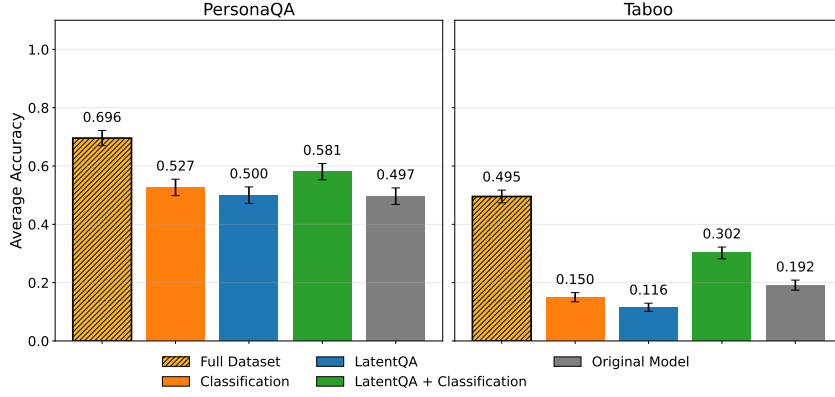


Figure 19: **Qwen3-8B results on PersonaQA and Taboo across training mixtures.** Accuracy of Activation Oracles trained with different datasets on two Qwen3-8B evaluations: recovering fine-tuned persona attributes (PersonaQA) and recovering a hidden secret word (Taboo-style secret-keeping). The untrained base model and SPQA-only AO perform poorly on both tasks, while adding classification and especially self-supervised context-prediction data yields large gains.

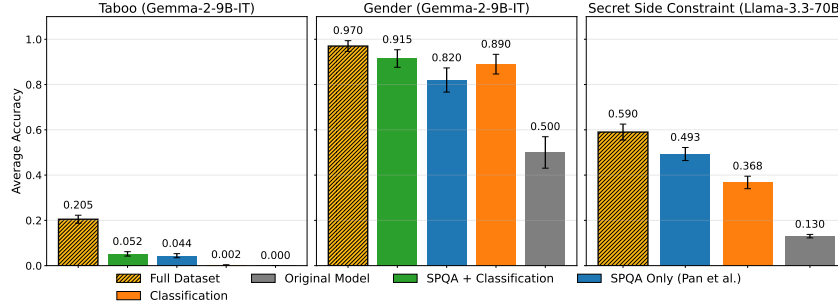


Figure 20: **Secret-keeping performance for different Activation Oracle training mixtures.** Accuracy on the three secret-keeping benchmarks: Taboo and Gender (Gemma-2-9B-IT) and Secret Side Constraint (Llama-3.3-70B), for the original model, single-task AOs, and the full Activation Oracle mixture. Across tasks, the full mixture achieves the best performance, but SPQA-only is already competitive: it clearly improves over the original model, and on Gemma secret-keeping tasks it matches or nearly matches the best white-box baselines.

C.7 DETAILED DATASET ABLATION RESULTS

In this section, we provide additional ablations on how the Activation Oracle training mixture affects performance on secret-keeping and related evaluations. Figure 19 focuses on Qwen3-8B, a setting where SPQA-only performs poorly and benefits substantially from adding classification and context-prediction data.

Figure 20 examines the effect of different training mixtures when evaluating the secret-keeping models.